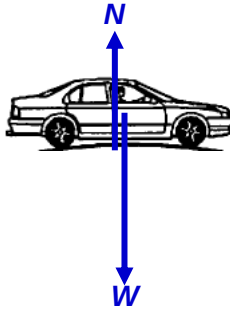


1	A car of mass 1700 kg travels over a curved hump in the road as shown in Fig. 1.1. The radius of curvature of the hump is 45 m.  Fig. 1.1		
	(a)	The speed of the car at the top of the hump is 19 m s⁻¹. Determine, for the car at the top of the hump,	
	(i)	the magnitude of the centripetal force acting on the car,	
		centripetal force = N	[1]
	L1	$F_c = \frac{mv^2}{r} = \frac{(1700)(19)^2}{45} = 13638 = 14000 \text{ N}$	A1
	(ii)	the magnitude of the normal contact force exerted by the road on the car.	
		normal contact force = N	[2]
	L2	Free body diagram:  The resultant force of the downward weight of the car W and the upward normal contact force exerted by the road N provide for the centripetal force required. $W - N = 13638$ $(1700)(9.81) - N = 13638$ $N = 3039 = 3000 \text{ N (magnitude)}$ Direction of N: vertically upwards	
	(b)	Determine the maximum speed v_{max} that the car can travel at without losing contact with the top of the hump. Explain your working.	
		$v_{max} = \text{ m s}^{-1}$	[3]
	L2	The resultant force of the downward weight of the car W and the upward normal contact force exerted by the road N provide for the centripetal force F_c required. Thus, $W - N = F_c$ $N = W - F_c$	B1

	For the car not to lose contact with the top of the hump, the normal contact force N with the road must be greater than zero, i.e. $N > 0$ $W - F_c > 0$ $mg - \frac{mv^2}{r} > 0$ $v < \sqrt{rg}$ $v < \sqrt{(45)(9.81)}$ $v_{\max} = 21.011 = 21 \text{ m s}^{-1}$	B1 A1
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[Total: 6]

2	A long, straight wire W carrying a direct current of 2.0 A flows in the direction as shown in Fig. 2.1
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